1 $f(ax) \Leftrightarrow \frac{1}{a}F(\frac{u}{a})$, where $a \in R$ and $a \neq 0$

1.1 Forward Direction: $f(ax) \Rightarrow \frac{1}{a}F(\frac{u}{a})$

$$\mathcal{F}\{f(ax)\} = \int_{-\infty}^{\infty} f(ax)e^{-j2\pi ux}dx \tag{1a}$$

Let $v = ax$, then $x = \frac{v}{a}$, which gives $dx = \frac{dv}{a}$:

$$\mathcal{F}\{f(ax)\} = \int_{-\infty}^{\infty} f(v)e^{-j2\pi u\frac{v}{a}}\frac{dv}{a} \tag{1b}$$

$$= \frac{1}{a}\int_{-\infty}^{\infty} f(v)e^{-j2\pi\frac{u}{a}v}dv \tag{1c}$$

$$= \frac{1}{a}F\left(\frac{u}{a}\right) \tag{1d}$$

1.2 Inverse Direction: $f(ax) \Leftarrow \frac{1}{a}F(\frac{u}{a})$

$$\mathcal{F}^{-1}\left\{\frac{1}{a}F\left(\frac{u}{a}\right)\right\} = \int_{-\infty}^{\infty}\frac{1}{a}F\left(\frac{u}{a}\right)e^{j2\pi ux}du \tag{2a}$$

Let $v = \frac{u}{a}$, then $u = av$, which gives $du = a\,dv$:

$$\mathcal{F}^{-1}\left\{\frac{1}{a}F\left(\frac{u}{a}\right)\right\} = \int_{-\infty}^{\infty}\frac{1}{a}F(v)e^{j2\pi avx}(a\,dv) \tag{2b}$$

$$= \int_{-\infty}^{\infty}F(v)e^{j2\pi avx}dv \tag{2c}$$

$$= f(ax) \tag{2d}$$

Thus, we have proven $f(ax) \Leftrightarrow \frac{1}{a}F(\frac{u}{a})$.

2 $f(x - x_0) \Leftrightarrow F(u)e^{-j2\pi ux_0}$

2.1 Forward Direction: $f(x - x_0) \Rightarrow F(u)e^{-j2\pi ux_0}$

$$\mathcal{F}\{f(x - x_0)\} = \int_{-\infty}^{\infty} f(x - x_0)e^{-j2\pi ux}dx \tag{3a}$$

Let $v = x - x_0$, then $x = v + x_0$, which gives $dx = dv$:

$$\mathcal{F}\{f(x - x_0)\} = \int_{-\infty}^{\infty} f(v)e^{-j2\pi u(v+x_0)}dv \tag{3b}$$

$$= \int_{-\infty}^{\infty} f(v)e^{-j2\pi uv}e^{-j2\pi ux_0}dv \tag{3c}$$

Since $e^{-j2\pi ux_0}$ is independent of the integration variable v, we can factor it out:

$$\mathcal{F}\{f(x - x_0)\} = e^{-j2\pi ux_0}\int_{-\infty}^{\infty} f(v)e^{-j2\pi uv}dv \tag{3d}$$

$$= F(u)e^{-j2\pi ux_0} \tag{3e}$$

2.2 Inverse Direction: $f(x - x_0) \Leftarrow F(u)e^{-j2\pi ux_0}$

$$\mathcal{F}^{-1}\{F(u)e^{-j2\pi ux_0}\} = \int_{-\infty}^{\infty} F(u)e^{-j2\pi ux_0}e^{j2\pi ux}du \tag{4a}$$

$$= \int_{-\infty}^{\infty} F(u)e^{j2\pi u(x-x_0)}du \tag{4b}$$

Let $v = x - x_0$, then $x = v + x_0$:

$$\mathcal{F}^{-1}\{F(u)e^{-j2\pi ux_0}\} = \int_{-\infty}^{\infty} F(u)e^{j2\pi uv}du \tag{4c}$$

$$= f(v) \tag{4d}$$

$$= f(x - x_0) \tag{4e}$$

Thus, we have proven $f(x - x_0) \Leftrightarrow F(u)e^{-j2\pi ux_0}$.